

Evaluation of a nurse-led disease management programme for chronic kidney disease: A randomized controlled trial

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ARTICLE INFO

Article history:

Received 23 January 2009

Received in revised form 23 March 2009

Accepted 5 July 2009

Keywords:

Chronic kidney disease management

Adherence

Kidney disease quality of life

Mixed skill nurse model

ABSTRACT

Background: Patients with end stage renal failure require dialysis and strict adherence to treatment plans to sustain life. However, non-adherence is a common and serious problem among patients with chronic kidney disease. There is a scarcity of studies in examining the effects of disease management programmes on patients with chronic kidney disease.

Objectives: This paper examines whether the study group receiving the disease management programme have better improvement than the control group, comparing outcomes at baseline (O1), at 7 weeks at the completion of the programme (O2) and at 13 weeks (O3).

Methods: This is a randomized controlled trial. The outcome measures were non-adherence in diet, fluid, dialysis and medication, quality of life, satisfaction, symptom control, complication control and health service utilisation.

Results: There was no significant difference between the control and study group for the baseline measures, except for sleep. Significant differences ($p < 0.05$) were found between the control and study group at O2 in the outcome measures of diet degree non-adherence, sleep, symptom, staff encouragement, overall health and satisfaction. Sustained effects at O3 were noted in the outcome measures of continuous ambulatory peritoneal dialysis (CAPD) non-adherence degree, sleep, symptom, and effect of kidney disease.

Conclusions: Many studies exploring chronic disease management have neglected the group with end stage renal failure and this study fills this gap. This study has employed an innovative model of skill mix using specialist and general nurses and demonstrated patient improvement in diet non-adherence, CAPD non-adherence, aspects of quality of life and satisfaction with care. Redesigning chronic disease management programmes helps to optimize the use of different levels of skills and resources to bring about positive outcomes.

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What is already known about the topic?

- Patients with end stage renal failure need to take an active role to follow a strict dialysis treatment regimens, medication and diet advice.
- Non-adherence is a common and serious problem among patients with chronic kidney disease.

- Disease management programmes have shown effects on adherence to treatment plan, symptom control, quality of life and satisfaction among patients with diabetes, heart failure and chronic obstructive pulmonary disease.

What this paper adds

- Many studies exploring chronic disease management have neglected the group with end stage renal failure and this study fills this gap.
- Effects achieved in this chronic kidney disease management model were improvement in non-adherence, aspects of quality of life and satisfaction with care.

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- This study has provided evidence to support an effective chronic kidney disease management model that uses skill mix using specialist and general nurses. This model is worth considering for other chronic disease groups.

1. Introduction

Patients with end-stage renal failure (ESRF) rely on artificial means, haemodialysis, continuous ambulatory peritoneal dialysis (CAPD) and the like, to sustain life. The treatment of CAPD demands that the patients take an active role in following a regular schedule of dialysis treatment regimens, strict medication and diet advice (Christensen, 2000). Literature suggests that non-adherence is a common and serious problem among patients with chronic kidney disease. Studies have been done to explore whether adherence can be enhanced among different chronically ill patient groups using a disease management approach. However, there is a scarcity of studies using the chronic kidney disease group as subjects.

1.1. Background

Statistics in 2006 showed that 506,256 individuals in the United States (United States Renal Data System, 2006), 107,825 in the United Kingdom (United Kingdom National Health Service Information Centre, 2006), and 83,189 in Hong Kong (Hong Kong Hospital Authority, 2006) suffered from ESRF. Concerns commonly identified among the ESRF patients are disruption of lifestyles, experience of continuing symptoms, stress and uncertainty of life (Polaschek, 2003). Hospital admission rates for renal patients range from 1.8–2.2 admissions per year (Lamping et al., 2000) to 4–7 times per year (Wong and Chung, 2006). When the progress of disease is not well controlled, patients suffer from poor quality of life (Lindqvist and Sjoden, 1998) and in turn make higher demands on the healthcare system (Lamping et al., 2000; Wong and Chung, 2006). Goals in the management of ESRF patients are therefore to improve symptom management, complication control and delay disease progression (Golper, 2001; Wong and Chung, 2006). Achieving these outcomes involves changing lifelong behaviours (Christensen and Ehlers, 2002) but this is not easy. A number of studies reported a non-adherence rate of 30–81.4%, mainly in fluid intake and dietary restrictions among the ESRF patients (Pang et al., 2001; Vlamincx et al., 2001; Christensen and Ehlers, 2002). One study found that one-third of peritoneal dialysis patients are believed to miss prescribed exchanges (Kutner, 2001).

The involvement of patients in adherence to a treatment regime is particularly important for ESRF patients because of the extreme dependence on artificial means to ensure treatment success and sustain life (Christensen and Ehlers, 2002). In the current disease management literature, it is recognized that single and episodic approach is not adequate to bring about sustainable adherence behaviour (Williams et al., 2008). A number of studies using different patient groups revealed that intensive

coaching and collaborative self-management with active participation of the client are important to bring about positive outcomes. Studies using groups of diabetic patients have shown that the intensive disease management approach is effective for glycaemic control (Norris et al., 2002; Wong et al., 2005a), screening for complications (Norris et al., 2002), and better blood monitoring and exercise adherence (Wong et al., 2005b). A systematic review of studies involving patients with heart failure supports disease management as a means to save costs and reduce hospitalizations (Riegel et al., 2002), improve symptoms and quality of life particularly related to physical functioning (Polaschek, 2003), improve adherence to treatment plans, and improve satisfaction (Riegel et al., 2002). Similar results were reported using patients with chronic obstructive pulmonary disease, including improved quality of life (Bourbeau et al., 2003), appropriate use of medication to control symptoms, improved psychological wellness (Egan et al., 2002) and reduced hospital admissions and emergency room visits (Bourbeau et al., 2003). For ESRF patients, some exploratory work illustrated that this specific group requires intensive care management to achieve desirable clinical outcomes (Holand, 1998; Nissenon, 2002) but no experimental study was conducted. This study is therefore launched to examine whether a disease management approach to managing the chronic kidney disease group would help enhance health outcomes similar to other disease groups. We focussed on the patients undergoing CAPD in this study.

1.2. Conceptual framework

This study adopts a disease management model that captures key features that have been identified in studies as important to support chronic care. The goals of chronic care are to control symptoms, prevent complications and promote a lifestyle that will delay disease progression (Rothman and Wagner, 2003). Wong et al. (2005b, 2008) have consolidated these features in a four-Cs model consisting of comprehensiveness, collaboration, coordination and continuity. Usually a nurse supported by a multi-disciplinary team will act as a case manager (Yu et al., 2006; Wong et al., 2008). Being comprehensive means that the case manager conducts a systematic assessment of patients' condition (Wong et al., 2008) including physical, psychological, and social, and is responsible for anticipating patients' health concerns (Hickey et al., 2000; Wong et al., 2008). The design of care support should include sustained follow-up to ensure early detection of complications, and patients' concerns and to reinforce adherence behaviours (Wagner et al., 2002). A collaborative process requires an appropriately organized healthcare network linked with available resources if patients need care from other providers (Wagner et al., 2002). The renal team in Hong Kong, like many other countries, is multi-disciplinary composed of nephrologists, nurses, social workers, dieticians and other health care providers by referrals. However, collaboration not only occurs between healthcare professionals, also between the providers and the patients. Patients need to be involved as partners and

active agents of care. The ultimate goal is to empower patients to assume responsibility for their own health (Von Korff et al., 1997). Coordination of services involves a delivery system designed to enable the case manager to operate across a spectrum of care in collaboration with physicians and the healthcare team in order to respond to patients' needs (Wong et al., 2008). The case manager has access to the expertise necessary to care for patients and his/her decision making is supported by the team (Wagner et al., 2002). Continuity emphasizes on care that is regular, active and has sustained follow-ups (Von Korff et al., 1997; Wagner et al., 2002).

2. The study

2.1. Design and research questions

This is a randomized controlled trial. 120 sets of computer-generated random numbers were used, and patients who fitted the criteria were randomized to the study or control group. Please refer to [Appendix A](#) for a brief outline of this study.

We asked whether the study group receiving the disease management programme was different from the control group in the outcome measures of non-adherence, quality of life, satisfaction, symptom and complication control and health care utilization.

2.2. Setting and participants

Subjects were recruited from two renal centres of a hospital cluster in Hong Kong. All patients who were on CAPD returning to the centres for clinical follow-up were invited to participate in the study.

The inclusion criteria were:

- Communicable
- Alert and oriented
- Could be contacted by telephone at home
- Lived in the hospital service area

The exclusion criteria were:

- On intermittent peritoneal dialysis or haemodialysis
- Old-age home residents

2.3. Ethical consideration

Information and explanation of the ethical observations of the study were provided to the subjects and they were asked to sign a consent form. The patients were free to withdraw from the study at anytime, and they were reassured that the decision whether or not to participate in the study would not prevent them from receiving the care that they would normally receive.

2.4. Intervention

The study group received both routine care and the intervention disease management programme. The control group received the routine care only, which included

instructions on medications and basic health advice. The design of the disease management programme included all the features of the four-Cs model. The programme was protocol-driven to govern the content and process of the intervention and the team was composed of renal nurses and general nurses. The renal nurses (ReNs) were nurses working in the study renal centres. The general nurses (GeNs) were nurses who were recruited from networks in the hospital and study university, and they served as volunteers in testing this innovative model. Each subject randomly assigned to the study group was followed up by a team composed of one ReN and one GeN. The ReN acted as a case manager to initiate and close the case and the GeN assisted the ReN in following up on the patient in the interim. We made this arrangement because we were informed by our previous study (Chow, 2005) that the renal nurses were pivotal in using their specialty knowledge and experience to make the initial assessment, identify problems and set mutual goals but the subsequent calls were mainly provided health advice and reinforced appropriate behaviours. Therefore in this study, we tried to optimize the use of human resources and arranged for the ReNs to make the initial and closing calls and the GeNs to provide calls in between. The ReNs were available for consultation at any time if the GeNs had questions. The duration of the disease management programme was 6 weeks following the work of Brooten et al. (2003) who showed that a period of 6 weeks was adequate to bring about therapeutic effects.

2.5. Training protocols for the renal nurses and general nurses

The ReNs went through a training programme of 24 h including theoretical input (8 h), case training and review (4 h) and guided readings on related content (12 h). The GeNs went through a training programme of 12 h, including theoretical input (8 h), case training and review (4 h). The contents of the protocols were validated by an expert panel, including a renal clinical nurse specialist, a renal physician, a renal nurse manager, a case manager and an academic. The theoretical content of the training of the ReNs included: (1) assessment using the Omaha system modified for renal patients, (2) arts and skills of telephone nursing, (3) setting mutual goals and health coaching, and (4) use of the disease management protocols, and (5) concepts of disease management, process and outcomes. The theoretical content of the training of the GeNs included items (1)–(4) same as above, and (5) chronic renal failure and the treatment modality of CAPD, and (6) health issues commonly encountered by patients undergoing CAPD and disease management. All nurses had to go through training using a simulated patient. One renal nurse and one academic listened to the audio-tapes and gave comments to each of the nurses.

2.6. Disease management protocols

The protocols were developed using the current literature and the expert opinions of the team in the renal

centres. The complete set of protocols was validated using the expert panel.

2.6.1. Pre-discharge assessment protocol

Before the patient was discharged from the clinic, the ReN conducted an initial assessment with the patient based on the Omaha framework. The Omaha system was originally derived from community health nursing practice and extensively field-tested in the United States (Martin, 2005). The research team has used the Omaha system for guiding the post-discharge care of general medical patients and found the system suitable for local use (Wong et al., 2008). The Omaha system encompasses four dimensions including environmental, psychosocial, physiological and health-related behaviours. We have replaced some items that were not relevant with items specific to patients with ESRF. Examples of removed items are 'pregnancy', 'ante-partum/postpartum' and examples of the added items are 'oedema' and 'fluid and diet restriction'. The final total number of health problems was 50.

2.6.2. Nurse-initiated telephone call protocols

The nurses would make phone calls to the patient every week for 6 weeks. There was no set time limit for each call but the calls were guided by a structured format including the following: (1) nurse's self-introduction and general address of the patient via asking about their overall health condition; (2) monitoring changes from the specific health concerns identified in the previous interaction, monitoring progress, providing health advice, reinforcing health self-management behaviours, and assessing the need for referral; (3) reviewing the health goals with the patient and setting mutual goals; (4) ending the telephone call using words of encouragement and arranging the time for the next call, if appropriate. We assigned the patients to the trained nurses by rotation and each nurse would not take more than two cases at any one time. The Omaha system continued to be used as a framework for assessment, intervention and documentation. The Omaha intervention scheme involved the four areas: health teaching and counselling, surveillance case management and treatment and procedures. The nurses in this study would mainly exercise the first three intervention areas since the delivery of treatment and procedures was not feasible in telephone follow-up. The first call was initiated by the ReNs within 72 h after the patient was discharged from the clinic. The subsequent calls were made by the GeNs every week for 4 weeks. The GeNs reinforced appropriate behaviours, identified potential complications and needs, and reviewed the mutual goal-setting. The ReNs would make the final call to review the health goals with the patient, provided advice if needed and closed the case.

2.7. Sample size

No previous studies had used a disease management intervention on ESRF patients undergoing CAPD, so we used the baseline data of disease complications and health-related behaviours in PD patients in Wong and Chung's (2006) study. The study showed that renal patients had a

21% improvement rate in terms of symptoms and complications between two renal nurse clinic visits. A sample size of 94 (47 case and 47 control) was derived based on one-way repeated measures using nQuery Advisor 4.0 (Elashoff, 2000). This design achieves 80% power to test between groups (study and control) with a 5% significance level and 0.29 effect size. We also validated the adequacy of this sample size by referencing a study which used a nurse telephone follow-up programme and adherence as a primary outcome for diabetic patients (Wong et al., 2005a). Based on blood glucose monitoring at 24 weeks, the effect size was 1.22 and only a sample of 12 subjects per group was needed to achieve a power of 80%.

2.8. Outcome measures

2.8.1. Non-adherence

The instrument was modified from the dialysis diet and fluid non-adherence questionnaire (DDFQ) developed in Belgium. Its criterion and construct validity were substantiated with a positive correlation between the DDFQ and biochemical and biological ratings (Vlaminck et al., 2001). It is a patient self-reported instrument measuring non-adherence behaviour (frequency and degree) with fluid and diet. We validated its relevance for local use using the expert panel and they recommended the inclusion of medication and dialysis non-adherence measures which were essential for ESRF patients. Also, the number of recall days was reduced to 7 instead of the original 14 days used in the DDFQ. The 7-day recall scale ensured that the most accurate information was obtained (Jerant et al., 2008). The final version of the questionnaire consisted of four domains (diet, fluid, dialysis, medication) each measuring the frequency (number of days of non-adherence) and degree of non-adherence (on a 5-point scale, 'no = 0' to 'very severe = 4'). The revised scale yielded good content validity (CVI = 1) and test–retest reliability among 15 renal patients for a 3-day interval with an intra-class correlation (ICC) of 0.98.

2.8.2. Quality of life

The short form of the Kidney Disease Quality of Life (KDQOL) was used. The instrument was originally developed by Hays et al. (1994) by adapting the SF-36 for generic health and incorporating renal disease-specific items. The overall internal consistency is around 0.70 (Hays et al., 1994; Martin and Thompson, 2000). The instrument was translated into Chinese and found to be valid for Chinese patients. The internal consistency ranged from 0.71–0.95 for the 11 kidney disease targeted scales, and 0.72–0.94 for the eight general health scales of the SF-36 (Li, 1998). The ICC test–retest reliability over 1 week among 15 renal patients was 0.81–0.96 for the different domains in the scale.

2.8.3. Satisfaction

The La Monica-Oberst Patient Satisfaction Scale was used (Munro et al., 1994). This is a 28-item instrument that measures the satisfaction of patients with accessibility/convenience of care, information giving, and

technical and professional competence. Translation validity was confirmed using forward and backward translation. The content validity for local use was confirmed by the expert panel. The internal consistency was good with a coefficient alpha of 0.91, and the test-retest ICC was 0.97.

2.8.4. Symptom and complication control

The items on symptom control included presence of oedema, existence of peritonitis, exit site condition and weight gain. The information provided by the subjects was validated against the documentation in their charts. The ICC inter-rater reliability was 0.93. The existence of complications was reflected in the monitoring of blood chemistry including urea, creatinine, sodium, potassium, phosphate and albumin (Vlaminck et al., 2001; Christensen and Ehlers, 2002; Wong and Chung, 2006).

2.8.5. Health service utilization

The health service utilization record included number of hospitalizations and emergency room visits. Data were extracted from the hospital computer systems.

2.9. Data collection

A trained research assistant (RA) helped collect the data by interviewing the patients using the structured questionnaires and retrieving clinical data from the charts. The RA was blind to group allocation and had no association with the clinical service.

Data collection happened at three time-points, at baseline (O1), at 7 weeks, i.e. the week after completion of the 6-week disease management programme (O2), and at 13 weeks, i.e. another 6 weeks after the completion of the disease management programme (O3).

2.10. Data analyses

Data were analysed using the Chi-square test and the independent *t*-test to compare the background characteristics of the study group and the control group. The independent *t*-test was used to compare the mean scores of each outcome measure between groups at each time period. We used one-way Analysis of Variance (ANOVA) repeated measures to determine between-groups (study vs. control) effects, within-groups (three times) effects and interaction (group $\times$ time) effects. All the data were analysed using SPSS 11.1 for Windows.

3. Results

3.1. Baseline data

There were more males (53.1%) than females (46.9%). The mean age was 62.4 years and most of them were married (63.3%). Their education level varied, 20.4% having no formal education and most being educated up to primary (41.8%) or secondary (33.7%) level. Only 9.2% were in employment and the rest were unemployed (11.2%), retired (53.1%) or homemakers (24.5%). The

majority of them had a place to live, in a self-owned flat (46.9%), public housing (39.8%), or rented accommodation (10.2%). Over one-third of the subjects reported that their financial situation was insufficient (24.5%) or extremely insufficient (6.1%), with the others stating that they had sufficient (60.2%) or more than sufficient (9.2%). The two groups shared similar co-morbidities profile with the top three diseases being diabetes (33.7%), hypertension (22.4%) and heart disease (16.3%). The average year on CAPD was 3.5 years and the mean number of daily exchange was 3. There was no significant difference in the background characteristics between the control and study groups.

3.2. Self-care non-adherence

The non-adherence level was equivalent between the study and control group at the baseline for all four domains of diet, fluid, medication and CAPD (Table 1). Significant findings were detected in a number of domains over time within groups and between groups. In the domain of diet days, both the study and control groups had improvement over time, and there was significant within-group improvement ($F = 13.65$, $p = 0.00$). There was no between-group difference. However, in the diet degree, there was significant between-group difference ($F = 4.19$, $p = 0.04$). The difference between the control and study groups was significant at O2 (study 0.49 vs. control 0.98, $t = 2.90$, $p = 0.00$) but not at O3 though the study group had a lower non-adherence degree score. In the domain of fluid intake, there was no significant difference noted in the frequency of fluid intake days. A significant difference was noted in the fluid intake degree within groups ($F = 8.68$, $p = 0.00$) and both the study and control groups seemed to improve over time. There was no significant difference for between-group comparison. There was no significant difference in the domain of medication days. For medication degree, the results were similar to those of fluid intake degree, that is, only within-group improvement was noted ($F = 8.45$, $p = 0.00$). Again, both the study and control groups seemed to have improved over time. As for the domain of CAPD, there was no significant difference noted in the CAPD days, but the CAPD non-adherence degree was significant within groups over time ($F = 4.68$, $p = 0.02$). There was no difference between groups at O2, but the study group made further progress at O3 and a significant difference was noted (study 0.00 vs. control 0.08, $t = 2.07$, $p = 0.04$). As for the interaction effect (group $\times$ time), no significant difference was noted for any of the four domains.

3.3. Quality of life

Table 2 shows that amongst the 20 aspects of the quality of life (QOL) measures, except for 'sleep', there was no significant difference between the two groups at the baseline measures. The study group had a better score for sleep than the control group at the baseline (study 60.5 vs. control 49.5, $t = -2.24$, $p = 0.03$) and maintained this significant difference at O2 (study 63.8 vs. control 47.9, $t = -3.40$, $p = 0.00$) and O3 (study 64.3 vs. control 48.1, $t = -3.81$, $p = 0.00$). The overall between-group difference

Table 1Non-adherence measures (control, $n = 49$; study, $n = 49$).

	O1 Mean (SD)	O2 Mean (SD)	O3 Mean (SD)	Between groups $F (p)$	Within groups $F (p)$	Interaction effect $F (p)$
Diet days				0.01 (0.92)	13.65 (0.00)	2.67 (0.09)
Study	1.97 (2.7)	0.74 (1.8)	0.69 (1.3)			
Control	1.49 (2.2)	1.09 (1.8)	0.92 (1.7)			
t -test (p)	−0.97 (0.33)	1.11 (0.27)	0.74 (0.46)			
Diet degree				4.19 (0.04)	17.44 (0.00)	1.65 (0.20)
Study	1.08 (1.1)	0.49 (0.7)	0.47 (0.7)			
Control	1.22 (1.0)	0.98 (1.0)	0.76 (1.0)			
t -test (p)	0.69 (0.49)	2.90 (0.00)	1.60 (0.11)			
Fluid intake days				0.08 (0.77)	2.60 (0.10)	0.96 (0.36)
Study	1.09 (2.3)	0.72 (1.4)	0.43 (1.1)			
Control	0.87 (2.0)	0.95 (1.9)	0.67 (1.6)			
t -test (p)	−0.51 (0.61)	0.67 (0.50)	0.88 (0.38)			
Fluid intake degree				1.74 (0.19)	8.68 (0.00)	0.68 (0.48)
Study	0.69 (0.8)	0.39 (0.7)	0.22 (0.5)			
Control	0.73 (1.0)	0.61 (0.8)	0.45 (0.9)			
t -test (p)	0.21 (0.83)	1.52 (0.13)	1.56 (0.12)			
Medication days				0.29 (0.59)	2.58 (0.09)	0.39 (0.63)
Study	0.27 (0.9)	0.14 (0.5)	0.12 (0.6)			
Control	0.43 (1.3)	0.12 (0.5)	0.18 (1.0)			
t -test (p)	0.71 (0.48)	−0.20 (0.84)	0.36 (0.72)			
Medication degree				0.70 (0.79)	8.45 (0.00)	0.32 (0.66)
Study	0.29 (0.6)	0.12 (0.4)	0.08 (0.3)			
Control	0.27 (0.6)	0.16 (0.4)	0.12 (0.3)			
t -test (p)	−0.17 (0.87)	0.50 (0.62)	0.66 (0.51)			
CAPD days				2.58 (0.11)	0.58 (0.53)	0.67 (0.49)
Study	0.02 (0.1)	0.02 (0.1)	0.00 (0.0)			
Control	0.20 (1.1)	0.02 (0.1)	0.16 (1.0)			
t -test (p)	1.18 (0.24)	0.00 (1.00)	1.13 (0.26)			
CAPD degree				1.82 (0.18)	4.68 (0.02)	0.19 (0.73)
Study	0.12 (0.3)	0.02 (0.1)	0.00 (0.0)			
Control	0.16 (0.5)	0.06 (0.2)	0.08 (0.3)			
t -test (p)	0.47 (0.64)	1.02 (0.31)	2.07 (0.04)			

Table 2KDQOL results (control, $n = 49$; study, $n = 49$).

	O1 Mean (SD)	O2 Mean (SD)	O3 Mean (SD)	Between groups $F (p)$	Within groups $F (p)$	Interaction effect $F (p)$
Symptom				5.12 (0.03)	2.17 (0.12)	1.88 (0.16)
Study	76.4 (14.0)	78.1 (15.2)	79.4 (12.9)			
Control	72.1 (16.0)	69.8 (16.8)	72.7 (16.5)			
t -test (p)	−1.39 (0.17)	−2.58 (0.01)	−2.23 (0.03)			
Effect of kidney disease				0.37 (0.55)	2.19 (0.13)	0.26 (0.72)
Study	68.9 (20.5)	71.7 (18.7)	72.0 (17.0)			
Control	67.3 (19.5)	68.3 (20.2)	70.9 (17.5)			
t -test (p)	−0.41 (0.69)	−0.85 (0.40)	−2.23 (0.03)			
Burden of kidney disease				1.15 (0.29)	1.07 (0.34)	0.39 (0.68)
Study	28.6 (21.2)	28.6 (21.2)	29.3 (19.0)			
Control	32.1 (22.5)	32.1 (22.5)	35.2 (23.9)			
t -test (p)	0.81 (0.42)	0.81 (0.42)	1.35 (0.18)			
Work status				2.73 (0.10)	1.19 (0.31)	0.25 (0.78)
Study	32.7 (26.1)	29.6 (28.7)	31.6 (30.1)			
Control	39.8 (28.8)	36.7 (28.5)	41.8 (27.7)			
t -test (p)	1.29 (0.20)	1.24 (0.22)	1.75 (0.08)			
Cognitive function				1.67 (0.20)	0.23 (0.77)	0.40 (0.64)
Study	75.1 (23.0)	74.8 (22.9)	74.7 (21.7)			
Control	78.1 (20.5)	80.4 (19.1)	80.7 (20.1)			
t -test (p)	0.68 (0.50)	1.31 (0.19)	1.42 (0.16)			

Table 2 (Continued)

	O1 Mean (SD)	O2 Mean (SD)	O3 Mean (SD)	Between groups <i>F</i> (<i>p</i>)	Within groups <i>F</i> (<i>p</i>)	Interaction effect <i>F</i> (<i>p</i>)
Quality of social interaction				0.01 (0.93)	0.71 (0.47)	2.01 (0.15)
Study	76.1 (20.2)	80.3 (18.8)	77.8 (20.4)			
Control	80.4 (17.9)	77.6 (20.3)	75.4 (21.3)			
<i>t</i> -test (<i>p</i>)	1.13 (0.26)	−0.69 (0.49)	−0.58 (0.56)			
Sexual function				0.14 (0.72)	0.22 (0.81)	0.22 (0.81)
Study	58.3 (37.6)	62.5 (30.6)	64.6 (30.0)			
Control	75.0 (0.0)	75.0 (0.0)	75.0 (0.0)			
<i>t</i> -test (<i>p</i>)	0.37 (0.72)	0.51 (0.63)	0.37 (0.73)			
Sleep				12.19 (0.00)	0.23 (0.76)	1.25 (0.29)
Study	60.5 (23.2)	63.8 (19.9)	64.3 (19.9)			
Control	49.5 (25.5)	47.9 (25.9)	48.1 (22.1)			
<i>t</i> -test (<i>p</i>)	−2.24 (0.03)	−3.40 (0.00)	−3.81 (0.00)			
Social support				1.27 (0.26)	1.81 (0.17)	0.05 (0.95)
Study	83.5 (17.1)	81.0 (21.1)	83.8 (16.2)			
Control	80.3 (20.9)	76.7 (23.1)	79.6 (21.0)			
<i>t</i> -test (<i>p</i>)	−0.84 (0.40)	−0.95 (0.34)	−1.12 (0.26)			
Staff encouragement				2.64 (0.11)	3.83 (0.02)	1.48 (0.23)
Study	85.2 (20.5)	90.8 (16.9)	90.1 (18.4)			
Control	81.4 (25.1)	81.9 (19.3)	88.0 (12.2)			
<i>t</i> -test (<i>p</i>)	−0.83 (0.41)	−2.44 (0.02)	−0.65 (0.52)			
Patient satisfaction				1.30 (0.26)	0.32 (0.71)	0.08 (0.91)
Study	79.3 (19.7)	79.3 (16.8)	79.9 (18.3)			
Control	82.0 (19.8)	83.0 (18.8)	84.0 (15.2)			
<i>t</i> -test (<i>p</i>)	0.68 (0.50)	1.04 (0.30)	1.20 (0.23)			
Physical functioning				0.02 (0.89)	0.60 (0.53)	0.69 (0.48)
Study	56.7 (28.7)	60.2 (27.1)	58.2 (27.5)			
Control	59.7 (29.9)	59.4 (28.5)	58.3 (27.7)			
<i>t</i> -test (<i>p</i>)	0.50 (0.62)	−0.15 (0.88)	0.02 (0.99)			
Role-physical				1.05 (0.31)	0.23 (0.79)	0.56 (0.58)
Study	28.6 (38.1)	29.8 (41.0)	25.0 (39.8)			
Control	35.0 (41.5)	36.9 (43.8)	37.5 (45.6)			
<i>t</i> -test (<i>p</i>)	0.73 (0.47)	0.76 (0.45)	1.32 (0.19)			
Pain				0.16 (0.69)	2.68 (0.08)	1.36 (0.26)
Study	68.7 (31.9)	73.1 (32.0)	76.5 (27.7)			
Control	71.0 (29.8)	67.2 (31.1)	73.5 (28.3)			
<i>t</i> -test (<i>p</i>)	0.37 (0.71)	−0.91 (0.36)	−0.52 (0.60)			
General health perception				0.64 (0.43)	7.00 (0.00)	0.49 (0.61)
Study	32.9 (21.9)	39.1 (23.8)	40.7 (24.2)			
Control	31.5 (16.4)	34.0 (20.4)	37.9 (24.5)			
<i>t</i> -test (<i>p</i>)	−0.34 (0.74)	−1.14 (0.26)	−0.58 (0.56)			
Emotional wellbeing				0.69 (0.41)	1.14 (0.72)	0.37 (0.66)
Study	66.9 (21.6)	69.6 (21.3)	70.8 (21.4)			
Control	65.5 (21.4)	65.2 (20.4)	67.0 (22.3)			
<i>t</i> -test (<i>p</i>)	−0.32 (0.75)	−1.03 (0.31)	−0.85 (0.40)			
Role-emotional				0.01 (0.93)	0.41 (0.64)	0.01 (0.98)
Study	53.1 (45.1)	55.8 (46.8)	57.1 (47.1)			
Control	53.7 (46.6)	57.1 (46.1)	57.1 (47.1)			
<i>t</i> -test (<i>p</i>)	0.07 (0.94)	0.14 (0.89)	0.00 (1.00)			
Social function				1.05 (0.31)	0.84 (0.44)	0.29 (0.75)
Study	54.6 (31.0)	58.9 (28.6)	55.6 (27.4)			
Control	62.2 (32.3)	63.0 (30.6)	59.7 (28.0)			
<i>t</i> -test (<i>p</i>)	1.20 (0.23)	0.68 (0.50)	0.73 (0.47)			
Energy/fatigue				0.50 (0.48)	2.38 (0.10)	0.66 (0.52)
Study	45.5 (20.5)	49.8 (22.2)	51.6 (22.4)			
Control	45.3 (22.6)	46.0 (20.0)	47.4 (23.6)			
<i>t</i> -test (<i>p</i>)	−0.05 (0.96)	−0.88 (0.38)	−0.90 (0.37)			
Overall health				2.76 (0.10)	10.78 (0.00)	2.20 (0.11)
Study	49.2 (13.8)	56.9 (16.2)	56.6 (16.5)			
Control	46.3 (21.1)	48.1 (21.2)	52.0 (19.1)			
<i>t</i> -test (<i>p</i>)	−0.79 (0.43)	−2.32 (0.02)	−1.27 (0.21)			

was significant ($F = 12.19$, $p = 0.00$) and the study group seems to have shown more improvement in sleep than the control group. We executed ANCOVA to account for the baseline difference and similar results were obtained. There was significant difference in O2 ($F = 6.25$, $p = 0.01$), O3 ($F = 8.98$, $p = 0.00$) and between-group difference ($F = 9.17$, $p = 0.00$). A significant between-group difference was also noted in 'symptom' ($F = 5.12$, $p = 0.03$), and the significant difference appeared both in O2 (study 78.1 vs. control 69.8, $t = -2.58$, $p = 0.01$) and O3 (study 79.4 vs. control 72.7, $t = -2.23$, $p = 0.03$). Besides sleep and symptom, no other aspect of QOL showed a between-group difference. A few items showed within-group differences, that is, both groups had improvements over time. They were staff encouragement ($F = 3.83$, $p = 0.02$), general health perception ($F = 7.00$, $p = 0.00$), and overall health ($F = 10.78$, $p = 0.00$). The study group had more improvement than the control group, and two of the items reached a statistical significance ($p < 0.05$). They were staff encouragement (study 90.8 vs. control 81.9, $t = -2.44$, $p = 0.02$) and overall health (study 56.9 vs. control 48.1, $t = -2.32$, $p = 0.02$) at O2. At O3, a significant difference was noted for effect of kidney disease (study 72.0 vs. control 70.9, $t = -2.23$, $p = 0.03$) but there was no between-group or within-group significance for this item.

3.4. Satisfaction

There was no significant difference between the control and study groups in the satisfaction level at baseline (study 71.04 vs. control 71.37, $t = 0.21$, $p = 0.84$). However, at O2 the study group had a significantly higher satisfaction score (study 77.12 vs. control 72.26, $t = -2.66$, $p = 0.01$), although the difference was insignificant at O3 (study 76.49 vs. control 74.04, $t = -1.18$, $p = 0.24$). The within group difference was significant ($F = 10.44$, $p = 0.00$) but that between groups was not ($F = 2.50$, $p = 0.12$).

3.5. Symptom control, complication control and health care utilization

There was no significant difference over time within groups or between groups for blood chemistry and symptom control, and all the readings were within acceptable range for CAPD patients. No difference was detected for the healthcare utilization rate either. The hospitalization rates at O2 and O3 were 4.1%, 20.4% (study) and 6.1%, 22.4% (control) respectively. The O2 and O3 visits to the emergency room were 4.1%, 16.3% (study) to 6.1%, 16.3% (control) respectively.

4. Discussion

This study has provided evidence to support an effective chronic kidney disease management model that uses a specialty-general nurse model. Results demonstrated patient improvement in non-adherence to self-care management in diet and CAPD, aspects of quality of life and satisfaction with care. Many studies exploring chronic disease management have neglected the group with ESRF, and this study fills this gap. It employed an innovative

model of skill mix using specialist and general nurses, which is worth considering for other chronic disease groups.

Many studies examining chronic disease management have focused on conditions such as diabetes (Ofman et al., 2004; Fisher et al., 2007), hypertension (Ofman et al., 2004; Williams et al., 2008), heart condition, and respiratory condition (Ofman et al., 2004). There is a paucity of studies using patients with chronic kidney disease as target subjects. The issues often encountered by patients with chronic conditions are non-adherence to treatment regimens (Ofman et al., 2004), which tends to be associated with the use of health services (Lamping et al., 2000; Wong and Chung, 2006) and quality of life (Polaschek, 2003). For the chronic kidney disease groups, Kutner (2001) noted that approximately one-third of patients on CAPD were non-compliant in prescribed exchanges. Christensen (2000) also revealed that 30% and 50% of patients with chronic kidney disease did not adhere to diet, fluid-intake, and medication regimens. Kammerer et al. (2007) argue that adherence requires patients to be active participants in establishing and achieving those goals in negotiation with healthcare providers. This study uses this contractual approach in establishing health goals to reinforce the health behaviours in the telephone follow-up.

This study did not show a change of blood chemistry with non-adherence behaviours. This concurs with another RCT study (Lee et al., 2006) which results showed that medication adherence was significantly improved but not the cholesterol level among 169 elderly patients for a 6-month follow-up. The degree of non-adherence was not translated into blood chemistry in both studies. The reasons were possibly related to a small sample size and short follow-up period. A study with a larger sample size ($n = 677$) and longer follow-up duration (3 years) was able to detect an association of 10% increase in non-adherence with increases of 0.14% in HbA1c and 4.9 mg/dl in cholesterol levels (Pladevall et al., 2004). In another study examining data of 14,930 renal patients, it was detected that the skipping of one or more haemodialysis sessions/month was associated with 7% increased phosphate, 13% of hospitalization and 30% of mortality rate (Saran et al., 2003).

In this study, we have conceptualized a model that uses the features that have been found to be effective for disease management in different chronic disease groups. These features are the essence of the four Cs which are comprehensiveness, collaboration, coordination, and continuity as expounded in the conceptual framework (Bourbeau et al., 2003; Norris et al., 2002; Wong et al., 2005a, 2005b). This study is one of only a few studies that apply a structured disease management programme among chronic kidney disease groups. What is interesting is that this model uses a mixed model of specialty and general nurse in the intervention and proves to be effective. The specialty nurse was the key person who conducted the initial assessment and planning and acted as a resource person for the general nurse throughout the disease management programme. Many of the disease management models tested tended to use advanced

practising nurses (APNs) such as nurse practitioners (Cumbie et al., 2004) and clinical nurse specialists (Wagner et al., 2002; Wong et al., 2005a) as case managers. These studies using APNs reported good outcomes, including improved quality of life (Delgado-Passler and McCaffrey, 2006), reduced re-admissions, decreased cost of providing health care (Naylor et al., 1999; Wong et al., 2005a), enhanced self-management behaviours (Wong et al., 2005a). It should be noted these studies mainly provided post-discharge follow-up care for patients after an acute episode of illness in the hospital. With the amount of evidence accumulated, it cannot be disputed that APNs can improve outcomes for patients with transitional health-care needs after hospitalization. This study, however, chose a group of chronic kidney disease patients who were relatively stable and returned to the clinics for follow-up. A mixed model using specialists and general nurses seems to be adequate to bring about positive effects for this type of patient who mainly needs support for health maintenance. Findings in this study concur with results from another study (Hanssen et al., 2007) that used non-specialty nurses to provide telephone follow-up intervention and proved to be effective in enhancing health behaviour and quality of life among a group of cardiac patients.

The competing resources in the contemporary health-care environment pose challenges to current disease management practice. In chronic disease management, individuals need close follow-up and proactive measures to meet needs and prevent complications (Von Korff et al., 1997; Wagner et al., 2002). The use of high-level professionals in all circumstances places demands on the already very busy healthcare providers. Our findings suggest that we can use a mixed model, employing nurses at the specialist level to lead the disease management programme and nurses at the general level to act in the supporting role for health maintenance. The complementary roles of each level of nurses were guided by set protocols and the specialist nurses acted as resource persons in case of deviations and the need for further actions. The collaboration of the patients in care is an important element in the disease management programme (Von Korff et al., 1997). The specialty nurse made the initiation and closing call and supported the general nurse who made a weekly call to the patients for the 6-week disease management programme. The team was able to establish a sustained relationship with the patients to provide health coaching and self-management support (Suter et al., 2008). In the United Kingdom, the *National Primary Care Development Team* (2009) also advocates the adoption of a multi-skilled multi-agency approach to ensure effective co-ordination in the care of people with chronic diseases as one of the key approaches in managing long-term conditions. In disease management programmes, there is always a concern regarding the sustained effects. In this study sustained effects were

achieved in some aspects but not all. These aspects included CAPD adherence behaviour, as well as sleep and symptom in quality of life. Findings suggest that the disease management intervention needs to be applied to the patients continuously so as to maintain a sustained effect (Von Korff et al., 1997). Regular reinforcement, being personal, and ongoing support for self-management are key features for effective disease management (Fisher et al., 2007).

5. Conclusion

Effective chronic disease management is important in maintaining quality of life and bringing about positive clinical outcomes. Disease management is a collaborative endeavour between patients and healthcare providers. Patients' involvement in self-care management contributes greatly to successful disease management, but they often need coaching and support from healthcare providers. Redesigning chronic disease management programmes helps to optimize the use of different levels of skills and resources to bring about positive outcomes. This study has provided evidence to support an effective chronic kidney disease management model that uses a specialty-general nurse model.

6. Limitations of the study

We acknowledge that there are a number of limitations to this study. The subjects were confined to renal patients receiving CAPD only and the results may not be able to be generalized to other chronic disease groups. With a small sample size and relatively short duration of follow-up, the effects of intervention could not be shown for outcome indicators, such as blood chemistry, that need a higher power or longer time to take effect.

Conflicts of interest

None declared.

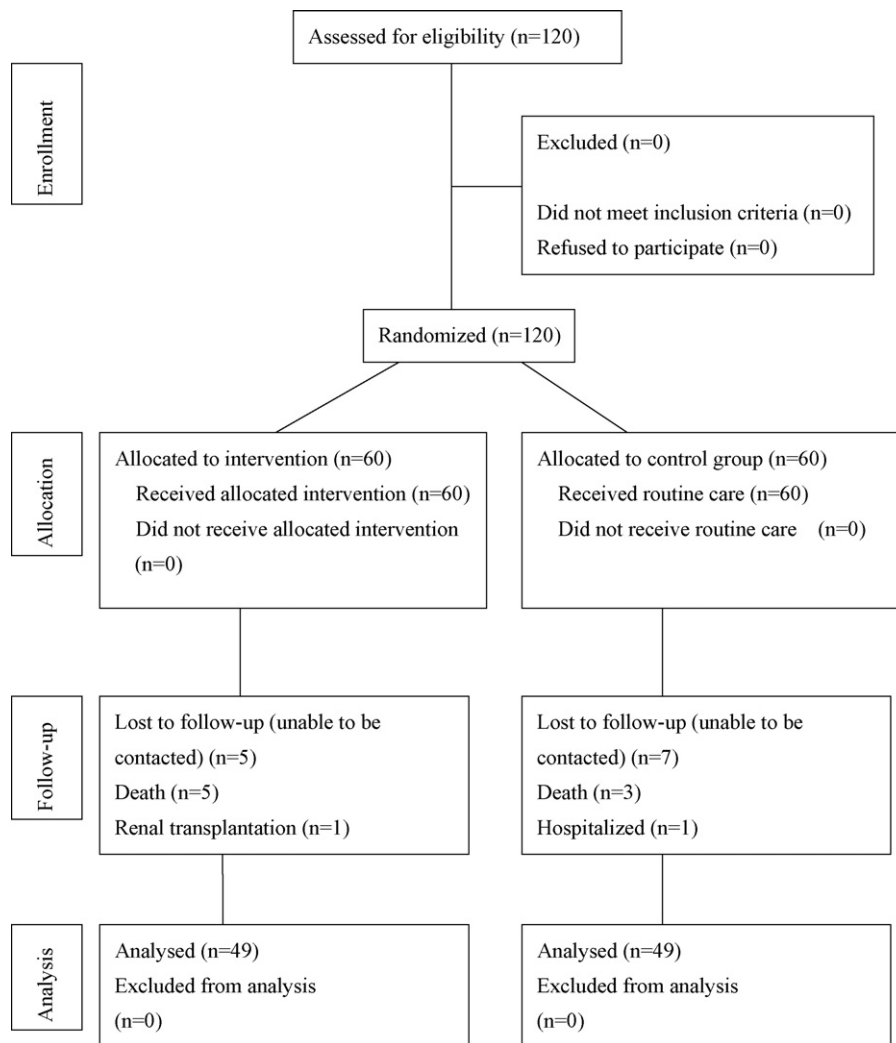
Acknowledgement

The authors would like to thank the renal nurses, general nurses and the patients who contributed to this study with the facilitation of Ms. Bonnie Tam, Nurse Specialist (Renal), Ms. Rebecca Yip, Ward Manager (Renal), and Ms. Helen Lee, Nursing Officer (Renal) of the study centres.

Funding: This research was funded by Research Grants Council of Hong Kong (PolyU 5435/05H).

Ethical approval: Ethical approval was given by the Hong Kong Polytechnic University (EA/03/229) and Queen Elizabeth Hospital (KC/KE-03-0015/ER-5).

Appendix A. Flow chart of the disease management programme study



References

- Bourbeau, J., Julien, M., Maltais, F., Rouleau, M., Beupre, A., Begin, R., Renzi, P., Nault, D., Borycki, E., Schwartzman, K., Singh, R., Collet, J., 2003. Reduction of hospital utilization in patients with chronic obstructive pulmonary disease: a disease-specific self-management intervention. *Archives of Internal Medicine* 163 (5), 585–591.
- Brooten, D., Youngblut, J.M., Deatrick, J., Naylor, M., York, R., 2003. Patient problems, advanced practice nurse (APN) interventions, time and contacts among five patient groups. *Journal of Nursing Scholarship* 35 (1), 73–79.
- Chow, S.K.Y. 2005. The effect of a nurse-led case management programme on patients undergoing peritoneal dialysis: a randomized control trial. PhD Thesis, School of Nursing, the Hong Kong Polytechnic University, Hong Kong.
- Christensen, A.J., 2000. Patient-by-treatment context interaction in chronic disease: a conceptual framework for the study of patient adherence. *Psychosomatic Medicine* 62 (3), 435–443.
- Christensen, A.J., Ehlers, S.L., 2002. Psychological factors in end-stage renal disease: an emerging context for behavioural medicine research. *Journal of Consulting & Clinical Psychology* 70 (3), 712–724.
- Cumbie, S., Conley, V., Burman, M., 2004. Advanced practice nursing model for comprehensive care with chronic illness: model for promoting process engagement. *Advances in Nursing Science* 27 (1), 70–80.
- Delgado-Passler, P., McCaffrey, R.N.D., 2006. The influences of post-discharge management by nurse practitioners on hospital readmission for heart failure. *Journal of the American Academy of Nurse Practitioners* 18 (4), 154–160.
- Egan, E., Clavarino, A., Burrige, L., Teuwen, M., White, E., 2002. A randomized control trial of nursing-based case management for patients with chronic obstructive pulmonary disease. *Lippincott's Case Management* 7 (5), 170–179.
- Elashoff, J.D., 2000. nQuery Advisor R 4.0. Statistical Solutions. Stonehill Corporate Center, Saugus.
- Fisher, E.B., Brownson, C.A., O'Toole, M.L., Anwuri, V.V., 2007. Ongoing follow-up and support for chronic disease management in the Robert Wood Johnson Foundation Diabetes Initiative. *Diabetes Educator* 33 (S6), 201–207.
- Golper, T., 2001. Patient education: can it maximize the success of therapy? *Nephrology Dialysis Transplantation* 16 (S7), 20–24.
- Hanssen, T.A., Nordrehaug, J.E., Eide, G.E., Hanestad, B.R., 2007. Improving outcomes after myocardial infarction: a randomized controlled trial evaluating effects of a telephone follow-up intervention. *European Journal of Cardiovascular Prevention & Rehabilitation* 14 (3), 429–437.

- Hays, R.D., Kallich, J.D., Mapes, D.L., Coons, S.J., Carter, W.M., 1994. Development of the kidney disease quality of life (KDQOL) instrument. *Quality of Life Research* 3 (5), 329–338.
- Hickey, M.L., Cook, F., Rossi, L.P., Connor, J., Dutkiewicz, C., McCabe, S.H., Fay, M., Lee, T.H., Fairchild, D.G., 2000. Effect of case managers with a general medical patient population. *Journal of Evaluation in Clinical Practice* 6 (1), 23–29.
- Holland, J., 1998. Integrating the role of the renal nurse case manager. *Nephrology News & Issues* 12 (1), 19–23.
- Hong Kong Hospital Authority, 2006. Hospital authority statistical report 2006–2007, Hong Kong.
- Jerant, A., DiMatteo, R., Arnsten, J., Moore-Hill, M.M., Franks, P., 2008. Self-report adherence measures in chronic illness: retest reliability and predictive validity. *Medical Care* 46 (11), 1134–1139.
- Kammerer, J., Garry, G., Hartigan, M., Carter, B., Erlich, L., 2007. Adherence in patients on dialysis: strategies for success. *Nephrology Nursing Journal* 34 (5), 479–486.
- Kutner, N.G., 2001. Improving compliance in dialysis patients: does anything work? *Seminars in Dialysis* 14 (5), 324–327.
- Lamping, D.L., Constantinovici, N., Roderick, P., Normand, C., Henderson, L., Harris, S., Brown, E., Gruen, R., Victor, C., 2000. Clinical outcomes, quality of life, and costs in the North Thames Dialysis Study of elderly people on dialysis: a prospective cohort study. *The Lancet* 356 (4), 1543–1550.
- Lee, J.K., Grace, K.A., Taylor, A.J., 2006. Effect of a pharmacy care program on medication adherence and persistence, blood pressure, and low-density lipoprotein cholesterol: a randomized controlled trial. *JAMA* 296 (21), 2563–2571.
- Li, C.M., 1998. Development of the Hong Kong Chinese version of the kidney disease quality of life short form (KDQOL-SFTM). Unpublished Master's Thesis, The Hong Kong Polytechnic University, Hong Kong.
- Lindqvist, R., Sjoden, P., 1998. Coping strategies and quality of life among patients on continuous ambulatory peritoneal dialysis (CAPD). *Journal of Advanced Nursing* 27 (2), 312–319.
- Martin, C.R., Thompson, D.R., 2000. Prediction of quality of life in patients with end-stage renal disease. *British Journal of Health Psychology* 5 (2), 41–55.
- Martin, K., 2005. *The Omaha System: A Key to Practice, Documentation, and Information Management*, 2nd ed. Elsevier Saunders, St. Louis.
- Munro, B.H., Jacobsen, B.S., Brooten, D.A., 1994. Re-examination of psychometric characteristics of the La Monica-Oberst patient satisfaction scale. *Research in Nursing & Health* 17 (3), 119–125.
- National Primary Care Development Team, 2009. Retrieved January 7, 2009 from <http://www.networks.nhs.uk/44.php>.
- Naylor, M., Brooten, D., Campbell, R., Jacobsen, B., Mezey, M., Pauly, M., Schwartz, S., 1999. Comprehensive discharge planning and home follow-up of hospitalized elders: a randomized clinical trial. *Journal of the American Medical Association* 281 (7), 613–620.
- Nissenon, A., 2002. Outcomes associated with a disease management program for end-stage renal disease. *Disease Management and Health Outcomes* 10 (2), 93–100.
- Norris, S.L., Nichols, P.J., Caspersen, C.J., Glasgow, R.E., Engelgau, M.M., Jack, L., Isham, G., Snyder, S.R., Carande-Kulis, V.G., Garfield, S., Briss, P., McCulloch, D., 2002. The effectiveness of disease and case management for people with diabetes: a systematic review. *American Journal of Preventive Medicine* 22 (4S), 15–38.
- Ofman, J.J., Badamgarav, E., Henning, J.M., Knight, K., Gano, A.D., Levan, R.K., Gur-Arie, S., Richards, M.S., Hasselblad, V., Weingarten, S.R., 2004. Does disease management improve clinical and economic outcomes in patients with chronic diseases? A systematic review. *The American Journal of Medicine* 117 (3), 182–192.
- Pladevall, M., Keoki, W.L., Potts, L.A., Divine, G., Xi, H., Lafata, J., 2004. Clinical outcomes and adherence to medications measured by claims data in patients with diabetes. *Diabetes Care* 27 (12), 2800–2805.
- Pang, S., Ip, W., Chang, A., 2001. Psychosocial correlates of fluid compliance among Chinese haemodialysis patients. *Journal of Advanced Nursing* 35 (3), 691–698.
- Polaschek, N., 2003. Living on dialysis: concerns of clients in a renal setting. *Journal of Advanced Nursing* 42 (4), 255–263.
- Riegel, B., Carlson, B., Kopp, A., LePetri, B., Glaser, D., Unger, A., 2002. Effects of a standardized nurse case-management telephone intervention on resource use in patients with chronic heart failure. *Archives of Internal Medicine* 162 (6), 705–712.
- Rothman, A.A., Wagner, E.H., 2003. Chronic illness management: what is the role of primary care? *Annals of Internal Medicine* 138 (3), 256–261.
- Saran, R., Bragg-Gresham, J.L., Rayer, H.C., Goodkin, D.A., Keen, M.L., vanDijk, P.C., Kurokawa, K., Piera, L., Saito, A., Fukuhara, S., Young, E.W., Held, P.J., Port, F.K., 2003. Nonadherence in hemodialysis: associations with mortality, hospitalization, and practice patterns in the DOPPS. *Kidney International* 64 (1), 254–262.
- Suter, P., Hennessey, B., Harrison, G., Fagan, M., Norman, B., Suter, N., 2008. Home-based chronic care: an expanded integrative model for home health professionals. *Home Healthcare Nurse* 26 (4), 222–228.
- United Kingdom National Health Service Information Centre, 2006. Retrieved December 5, 2008 from <http://www.hesonline.nhs.uk>.
- United States Renal Data System, 2006. Reference tables on end-stage renal disease. Retrieved December 5, 2008 from <http://www.usrds.org/adr.htm>.
- Vlaminck, H., Maes, B., Jacobs, A., Reyntjens, S., Evers, G., 2001. The dialysis diet and fluid non-adherence questionnaire: validity testing of a self-report instrument for clinical practice. *Journal of Clinical Nursing* 10 (5), 707–714.
- Von Korff, M., Gruman, J., Schaefer, J., Curry, S.J., Wagner, E.H., 1997. Collaborative management of chronic illness. *Annals of Internal Medicine* 127 (12), 1097–1102.
- Wagner, E.H., Davis, C.M.N., Schaefer, J., Von Korff, M., Austin, B., 2002. A survey of leading chronic disease management programs: are they consistent with the literature? *Journal of Nursing Care Quality* 16 (2), 67–80.
- Williams, A., Manias, E., Walker, R., 2008. Interventions to improve medication adherence in people with multiple chronic conditions: a systematic review. *Journal of Advanced Nursing* 63 (2), 132–143.
- Wong, F.K.Y., Mok, M., Chan, T., Tsang, M.W., 2005a. Nurse follow-up of patients with diabetes: randomized controlled trial. *Journal of Advanced Nursing* 50 (4), 391–402.
- Wong, S., Wong, F.K.Y., Chan, T., 2005b. Effects of nurse-initiated telephone follow-up on self-efficacy among patients with chronic obstructive pulmonary disease. *Journal of Advanced Nursing* 49 (2), 210–222.
- Wong, F.K.Y., Chung, L., 2006. Establishing a model for nurse-led clinic: structure, process and outcome. *Journal of Advanced Nursing* 53 (3), 358–369.
- Wong, F.K.Y., Chow, S., Chung, L., Chang, K., Chan, T., Lee, W.M., Lee, R., 2008. Can home visits help reduce hospital readmissions? A randomized controlled trial. *Journal of Advanced Nursing* 62 (5), 585–595.
- Yu, D., Thompson, D.R., Lee, D.T.F., 2006. Disease management programmes for older people with heart failure: crucial characteristics which improve post-discharge outcomes. *European Heart Journal* 27 (5), 596–612.